1. Degree-two image-closed fiber calculus for the n=83 cover

2026-11-01 · Exact closed-point audit and strict source-rho gate

The first natural higher-degree image-compatible design is rejected before relation collection. For the cubic cover

$$E : y^2 + xy = x^3 + 1, \quad C : v^2 + u^2 v = u^7 + u, \quad \varphi(u, v) = (u^3, uv + 1),$$

every requested odd-degree generic rational fiber has one degree-one and one degree-two closed component. The general Jacobian norm maps them to P and $[2]P$. Their projected relation is therefore only $[3]P = P + [2]P$, a scalar-multiplication identity with independent logarithmic rank zero.

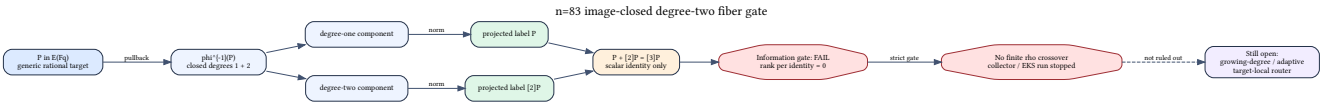


Figure 1: The exact component norm fails the information gate, so the collector and EKS timing stage are stopped.

1.1. Exact evidence

Fixture	Degrees	Projected labels	Control or conclusion
`n=9,13,19,23,83`	`[1,2]`	$P, [2]P$	Requested odd-degree cases
`n=2`	`[1,1,1]`	P, P, P	Positive split control
`n=83, (0,1)`	reduced 1	P	Ramification index 3 retained separately
Pinned order-r point	`[1,2]`	distinct $P, [2]P$	Target subgroup check

The native certificate factors each reduced Mumford support, reconstructs it, pushes every irreducible factor using the residue-field norm, sums the component images, and compares the result with the general arbitrary-class pushforward. For each point-class pullback it checks that a degree-d component maps to $[d]P$.

For the pinned generator

$$r = 241785163923079621668568 9,$$

the replay checks $[r]P = O$, the quadratic component maps to $[2]P$, and this label is neither O nor P. Because r is odd prime, doubling is injective on the subgroup. The apparent extra closed point creates no projected collision or compression.

The even-degree split case catches an implementation that always reports a 1+2 factorization. The exceptional point catches confusion between reduced support degree and scheme-theoretic ramification multiplicity.

1.2. Information and complete-cost gates

Measure after projected-label merging	Result
Distinct labels per generic odd fiber	2
Projected collision rate	0
Useful relation probability	0
Independent rank per identity	0
Admitted matrix rows / columns	0 / 0
Admitted matrix memory	0 bytes
Individual-log success	false

If $P = [a]G$ is source-labelled, the two component labels already have scalars a and $2a$. If P is the unknown target Q, the identity $[3]Q = Q + [2]Q$ reveals no logarithm. This is why the zero matrix entries mean "early stop," not a zero-cost completed attack.

All asymptotic accounting uses the actual order r. Source rho costs $r^{\frac{1}{2}}$. Projecting ordinary ambient genus-three relations retains the approximately $q^{\frac{4}{3}}$ collection scale: at $q = 2^{83}$ its exponent is 110.666… bits, about 70.17 bits above source rho. The independent fixed-bounded-branch gate is $\Omega(r)$ before linear algebra. Neither route predicts a finite crossover, and neither is admitted as a fallback.

1.3. Scope and remaining obligation

The rejection is deliberately narrow. It closes the degree-two component of a rational fiber and prevents raw upstairs multiplicity from being counted as projected rank. It does not prove that every higher-degree image-compatible calculus is impossible.

The remaining unclassified object is a compact growing-degree or adaptive target-local router. Before implementation it must name and cost:

- the target-local relation search and its useful probability;
- distinct projected columns and verified independent rank;
- sparse matrix dimensions, memory, and solve work;
- blind individual-log descent and final verification;
- a finite crossover against the strong source rho implementation.

The Lean theorem `degree_one_plus_degree_two` proves the final group identity in any additive monoid. It does not formalize the finite-field factorization or Rust norm path; those are covered by the native certificate and controls.

1.4. Sources

- Gaudry, *Index calculus for abelian varieties of small dimension and the elliptic curve discrete logarithm problem*, JSC 44 (2009), DOI.
- Gaudry, Thomé, Thériault, Diem, *A double large prime variation for small genus hyperelliptic index calculus*, ePrint 2004/153.
- Diem, *Index calculus with double large prime variation for curves of small genus with cyclic class group*, arXiv:math/0606607.
- Repository screens `n83_genus3_image_solver_20261031`, `n83_jacobian_projector_20261031`, and the P1546 projected-smoothness gate.

This report makes no wall-clock, throughput, attack, security, or speedup claim. The frozen gate rejected the candidate before any EKS timing run.